

## 1. Administrative & Study Identification

**Full Study Title:** A Randomized, Double-Blind, Placebo-Controlled, 52-Week Phase II Study to Evaluate the Efficacy of Intravenous R07046015/Prasinezumab (PRX002) in Participants With Early Parkinson's Disease With a Multi-Year All-Participants-on-Treatment Extension **Short Title/Acronym:** PASADENA Study **Protocol Number:** BP39529 **NCT Number:** NCT03100149 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Prasinezumab (R07046015 / PRX002) **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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## 2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the efficacy of intravenous prasinezumab (R07046015) versus placebo over 52 weeks on the progression of global motor and non-motor symptoms in participants with early Parkinson's disease (PD). **Secondary Objectives:** To assess the effect of prasinezumab on motor progression (measured by MDS-UPDRS Part III), time to clinically meaningful worsening of motor symptoms, imaging biomarkers (DaT-SPECT), safety, and pharmacokinetics.

**2.2 Background & Rationale** Aggregated alpha-synuclein plays a critical role in the pathogenesis and progression of Parkinson's disease. Prasinezumab is a humanized monoclonal antibody designed to selectively target and clear toxic alpha-synuclein aggregates, potentially preventing cell-to-cell transmission. This study aims to determine whether clearing these aggregates can result in clinically meaningful slowing of disease progression in patients with early-stage PD who are not yet dependent on dopaminergic therapies.

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## 3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 (Placebo : Prasinezumab Low Dose : Prasinezumab High Dose)

**3.2 Planned Enrollment & Duration Target \$N\$:** 316 evaluable patients **Number of Sites:** Approximately 60 sites across the United States, France, Austria, Germany, and Spain. **Estimated Study Start:** June 2017 **Estimated Primary Completion:** 52-week Part 1 completed (~2020), with extension phases active up to estimated Dec 2031.

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age 40 to 80 years. 2. Diagnosis of idiopathic PD for  $\leq 2$  years at screening, presenting with bradykinesia plus resting tremor or rigidity (Hoehn and Yahr Stage I or II). 3. Screening brain DaT-SPECT consistent with Parkinson's Disease. 4. Untreated or on a stable regimen of monoamine oxidase B (MAO-B) inhibitors.

**4.2 Exclusion Criteria** 1. Prior treatment with dopaminergic medication (e.g., levodopa or dopamine agonists) for  $> 60$  days, or lack of observable response to a sufficiently high dose of levodopa. 2. Concomitant use of COMT inhibitors, amantadine, anticholinergics, or specific antidepressants (fluoxetine, fluvoxamine). 3. Any significant cardiovascular condition or contraindications to obtaining a brain MRI.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Participants are randomized to one of three arms: Placebo, Prasinezumab Low Dose (1500 mg), or Prasinezumab High Dose (4500 mg for body weight  $\geq 65$  kg, or 3500 mg for  $< 65$  kg). **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** 52 weeks (Part 1 double-blind), followed by a 52-week blinded dose extension (Part 2), and a subsequent multi-year open-label extension (Part 3).

**5.2 Concomitant & Prohibited Therapy Permitted:** Stable dosing of MAO-B inhibitors; stable anti-depressants (excluding fluoxetine/fluvoxamine). **Prohibited:** Levodopa or dopamine agonists (unless disease progression necessitates rescue during the study), COMT inhibitors, and investigational PD vaccines.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures                                                |
|----------|---------------|--------------------------------------------------------------------------|
| 1        | -28 to 0      | Informed Consent, Medical History, DaT-SPECT, MRI                        |
| 2        | 0             | Baseline                                                                 |
| 3        | 0             | Randomization, First IV Infusion, MDS-UPDRS                              |
| 4        | 4             | Interim                                                                  |
| 5        | 4             | Every 4 Weeks                                                            |
| 6        | 4             | IV Infusion of IP/Placebo, Safety Labs, Vitals, AE assessment            |
| 7        | 52            | End of Part 1                                                            |
| 8        | 52            | Primary Endpoint Assessment (MDS-UPDRS), DaT-SPECT, Transition to Part 2 |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Change from baseline to Week 52 in the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) Total Score (Parts I, II, and III). **Measurement Method:** Clinical neurological examination by certified raters.

**7.2 Secondary & Exploratory Endpoints** 1. Change from baseline in MDS-UPDRS Part III (motor examination) score. 2. Time to clinically meaningful worsening of motor symptoms. 3. Change in imaging biomarkers (DaT-SPECT).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard collection via Q4W onsite visits, with special attention to infusion-related reactions.

**Serious Adverse Events (SAEs):** Routine reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed by the sponsor.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for all subjects receiving at least one dose of study medication. **Sample Size Justification:** Approximately 316 participants (randomized 1:1:1) to achieve 80% power to detect a clinically meaningful difference in the MDS-UPDRS progression rate between the pooled prasinezumab arms and placebo. **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) for continuous efficacy endpoints.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

**Monitoring Plan:** Standard onsite and remote monitoring visits to assure data integrity and protocol adherence. **Audit Trail:** eCRF locking, centralized DaT-SPECT and MRI image reads, and validated data clarification mechanisms.

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## 11. Study Identifiers & Governance

**Primary ID:** BP39529 **Registry Number:** NCT03100149 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** PASADENA Study **Official Regulatory Title:** A Randomized, Double-Blind, Placebo-Controlled, 52-Week Phase II Study to Evaluate the Efficacy of Intravenous R07046015/Prasinezumab (PRX002) in Participants With Early Parkinson's Disease With a 6-Year All-Participants-on-Treatment Extension

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## 12. Clinical Framework (Parkinson's Specific)

**Primary Condition:** Early Parkinson's Disease **Diagnosis Threshold:** Diagnosis  $\leq$  2\$ years, Hoehn and Yahr Stage I-II, DaT-SPECT

consistent with PD **Medical Methodology:** Targeted Monoclonal Antibody Therapy against  $\alpha$ -synuclein **Optimization Target:** Slowing of global and motor disease progression

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### 13. Study Procedures & Methodology

**Management Pathway:** Intravenous Monoclonal Antibody Infusion  
**Education Protocol (HCP):** MDS-UPDRS standardization and DaT-SPECT imaging protocol training  
**Education Protocol (Patient):** Informed consent regarding infusion expectations, extension phases, and blinded dose transitions  
**Quality Audit Mechanism:** Central reading for DaT-SPECT and MRI; independent rating validations for UPDRS

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### 14. Observation & Follow-Up Schedule

**Total Duration:** 52 weeks (Part 1) + 52 weeks (Part 2) + Multi-year Extension (Part 3) + 12 weeks safety follow-up  
**Onsite Visit Cadence:** Every 4 weeks (Q4W) for infusions and evaluations  
**Remote Monitoring Frequency:** Intermittent per site procedures  
**Checklist Review Interval:** Full clinical and MDS-UPDRS assessment typically at baseline, intermediate waypoints, and Week 52

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### 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |